

CSC11006 - INTRODUCTION TO CLOUD COMPUTING SERVICES

PRACTICE

LAB: Enable BigQuery sandbox

I. General information

ID: LAB3
Period: 30 minutes
Deadline:

II. Outcome:

This lab will adapt to these following course's outcomes:

- G2.1
- G4.1

III. Describe:

The BigQuery sandbox allows you to explore a limited set of BigQuery features for free, helping you determine if it meets your needs. You can use BigQuery without entering credit card details or setting up a billing account. If you already have a billing account, you can still access BigQuery at no cost within the free usage tier

1. Enable BigQuery sandbox:

1.1. Sign in to Google Cloud Console: Go to Google Cloud Console.

The Google Cloud console is the graphical interface that you use to create and manage BigQuery resources and to run SQL queries.

Authenticate with your Google Account, or create a new one. On the welcome page, do the following:

- For Country, select your country.
- For Terms of Service, select the checkbox if you agree to the terms of service.
- Optional: If you are asked about email updates, select the checkbox if you want to receive email updates.

 Try Google Cloud for free

Step 1 of 2 Account Information



Cloud CTL

[Switch account](#)

Country

Vietnam

Email updates

- ☐ I would like to receive periodic emails on news, product updates and special offers from Google Cloud and Google Cloud Partners.

By using this application, you agree to the [Google Cloud Platform](#), [Supplemental Free Trial](#), and [any applicable services and APIs](#) Terms of Service.

Agree & continue

Access to all Google Cloud products

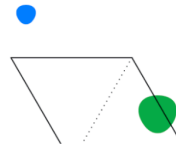
Get everything you need to build and run your apps, websites and services, including Firebase and the Google Maps API.

\$300 in free credit

Try Google Cloud with \$300 in credit to spend over the next 90 days.

No automatic charges

You only start paying if you decide to activate a full, pay-as-you-go account or choose to prepay. You'll keep any remaining free credit.



- Click Agree and continue.

Try Google Cloud for free

Step 2 of 2 Verify Your Identity

We use payment information to verify your identity and protect our services by spammers and bots. **You won't be charged** until you activate your full, pay-as-you-go account or choose to prepay.

Payments profile

Create new payments profile

Your payment information is saved in a payments profile linked to your Google Account and shared across Google services. [Learn more about payments profile](#)

Tax information (optional)

Add tax information

Please complete the previous sections before continuing.

Payment method

Add payment method

Please complete the previous sections before continuing.

Start free

Create a payments profile

Only **Organization** profiles can have multiple users. If you select an **Individual** profile, you agree that use of your profile is for your trade, business, craft, or profession. In some countries, this selection affects your tax options. Your profile type can't be changed after signing up. [Learn more about payments profile](#)

Profile type

Individual

Legal name

Cloud CTL

Street address

Nguyen Van Cu

Apt, suite, etc. (optional)

City

HCM

Province

Ho Chi Minh City

Postal code

700000

Cancel Create

You can skip this step and access BigQuery sandbox via the link:

<https://console.cloud.google.com/bigquery>

Items on the BigQuery sandbox welcome page.

Start your Free Trial with \$300 in credit. Don't worry—you won't be charged if you run out of credits. [Learn more](#)

Google Cloud My First Project Search (I) for resources, docs, products, and more Search

SANDBOX Set up billing to upgrade to the full BigQuery experience. [Learn more](#)

Explorer + ADD IK

Search BigQuery resources

Show started only

enhanced-gsmo-452302 v1

Welcome to BigQuery Studio!

Create new

SQL QUERY PYTHON NOTEBOOK DATA CANNAS DATA PREPARATION WORKFLOW

Try with sample data

Try the Google Trends Demo Query

This simple query generates the top search terms for Google Trends.

OPEN QUERY VIEW DATASET

Add your own data

Local file

Upload a local file

LAUNCH THIS GUIDE

Google Drive

Google storage service

LAUNCH THIS GUIDE

Google object storage service

LAUNCH THIS GUIDE

Summary

Nothing currently selected

Job history

Refresh

Welcome to BigQuery in the Cloud Console

New to the BigQuery UI?

The BigQuery UI helps you complete tasks like running queries, loading data, and even creating and training ML models. Check out the BigQuery [quickstart guide](#) to learn how to start performing data analysis on Google Cloud.

Learn about new features

New improvements and updates are constantly on the way. We recommend periodically checking our [release notes](#) to stay up to date on what's new.

DONE

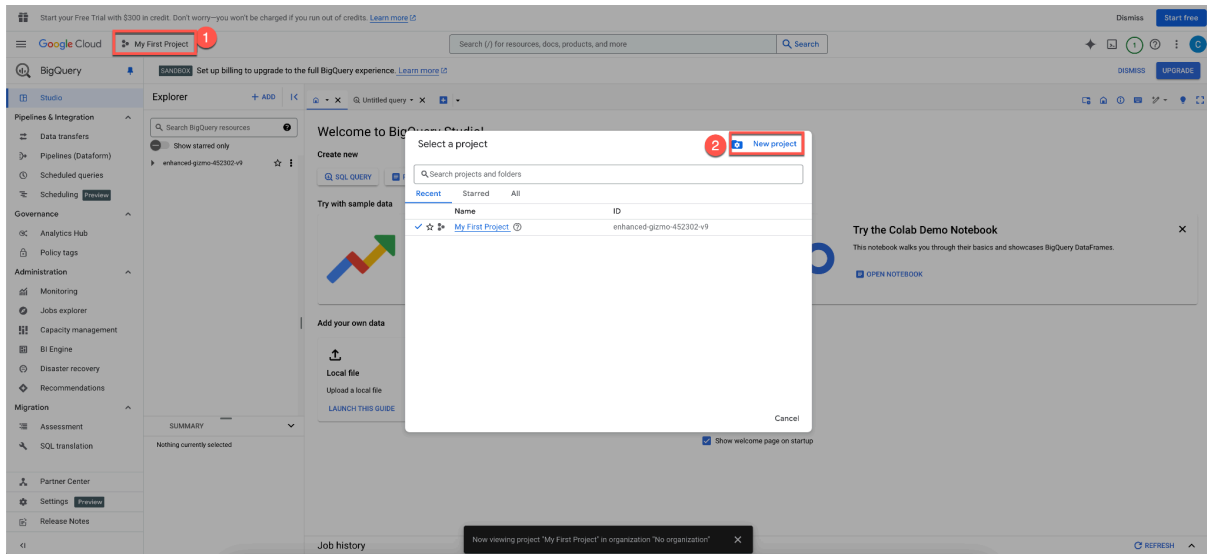
Try the Colab Demo Notebook

This notebook walks you through the basics and showcases BigQuery DataFrames.

OPEN NOTEBOOK

Show welcome page on startup

1.2 Select or create a project: If you haven't already, either create a new project or select an existing one from the project dropdown at the top of the page.

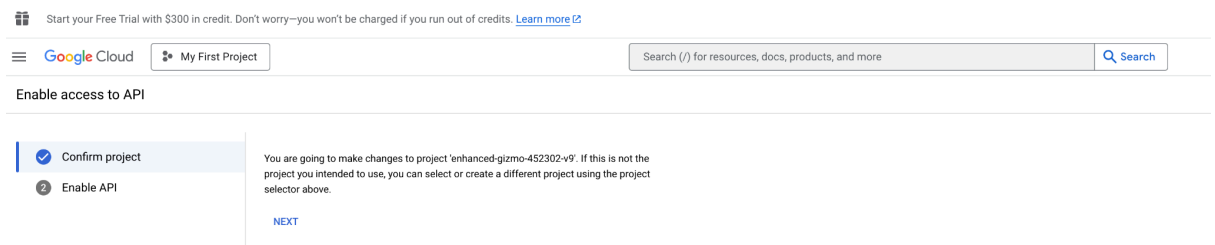


On the New Project page, do the following:

- For Project name, enter a name for your project.
- For Organization, select an organization or select No organization if you are not part of one. Managed accounts, such as those associated with academic institutions, must select an organization.
- If you are asked to select a Location, click Browse and select a location for your project.
- Click Create. You are redirected back to the BigQuery page in the Google Cloud console.
- BigQuery sandbox project creation page.

1.3 Enable BigQuery API:

- In the left sidebar, go to the "Navigation Menu" and select **BigQuery** under the "Big Data" section.
- If prompted, enable the **BigQuery API** for your project.



 Start your Free Trial with \$300 in credit. Don't worry—you won't be charged if you run out of credits. [Learn more](#)

Enable access to API

- ✓ Confirm project
- 2 Enable API

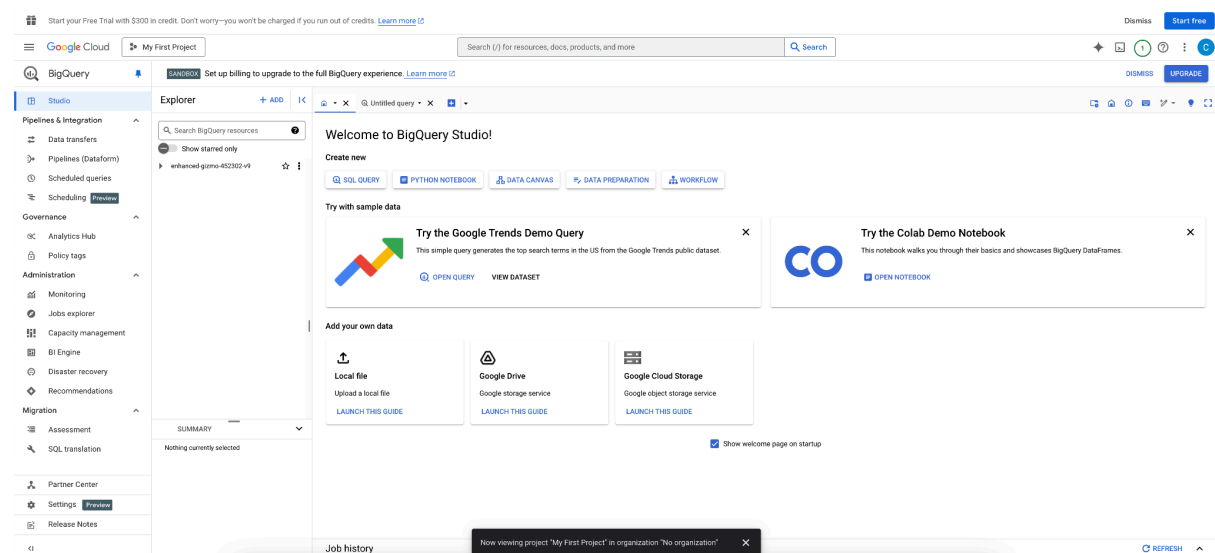
You are about to enable 'BigQuery API'.

ENABLE

1.4 Access the BigQuery Sandbox:

- BigQuery Sandbox is automatically available to any Google Cloud account without needing billing to be enabled. As long as you are within the free-tier usage limits (1 TB of querying and 10 GB of storage), you can use it.
- If you want to make sure you're using the sandbox, navigate to the **BigQuery console**, and you should see the "Sandbox" label under your project if it is using the free tier.

You have successfully enabled the BigQuery sandbox. A BigQuery sandbox notice is now displayed on the BigQuery page:



The confirmation notice provides the option to upgrade to the full BigQuery experience.

1.5 Start Using BigQuery Sandbox:

- Start running SQL queries and using BigQuery's features within the sandbox limits (1 TB per month for queries and 10 GB of storage).
- Note that if you exceed these limits, you'll be prompted to set up billing and switch to the standard BigQuery plan.

- With the BigQuery sandbox now enabled, we recommend that you try to query a public dataset with the Google Cloud console quickstart next. You can also try a similar quickstart for the `bq` command-line tool, which is useful for many administrative tasks, or a similar quickstart for the client libraries, which let you use BigQuery with programming languages like Java and Python.

2. Upgrade from the BigQuery sandbox

The BigQuery sandbox lets you explore limited BigQuery capabilities at no cost. When you are ready to increase your storage and query capabilities, upgrade from the BigQuery sandbox.

To upgrade, do the following:

- Enable billing for your project.
-
- Explore BigQuery editions and determine the pricing model that is right for you.
-
- Once you have upgraded from the BigQuery sandbox, you should update the default expiration times for your BigQuery resources such as tables, views, and partitions.

3. Limitations

The BigQuery sandbox is subject to the following limits:

- All BigQuery quotas and limits apply.
- You are granted the same free usage limits as the BigQuery free tier, including 10 GB of active storage and 1 TB of processed query data each month.
- All BigQuery datasets have a default table expiration time, and all tables, views, and partitions automatically expire after 60 days.
- The BigQuery sandbox does not support several BigQuery features, including the following:
 - Streaming data
 - Data manipulation language (DML) statements
 - BigQuery Data Transfer Service